geobard · turn GeoJSON into prose

OPEN-SOURCE · FASTAPI · OPENAI-COMPATIBLE

Turn GeoJSON into prose.

A small service that reads a map scene and writes what you'd see standing in it.

github.com/openfantasymap/geobard

A map is data. A place is a feeling.

A `FeatureCollection` is precise and machine-readable — and says nothing about **what it's like to be there.**

- Coordinates, types, names, geometry
- No "you're standing at the edge of a market square at first light"
- Players, agents, and readers want the **place**, not the table

geobard is the thin layer that closes that gap.

One small service. Three jobs.

geobard is a thin, well-prompted layer between **your map data** and **a language model.**

👁️ Window view

Describe the scene as if you stand in it. Human language only.

`POST /narrate/window`

💬 Ask the scene

Answer a question grounded in what's actually present.

`POST /narrate/prompt`

🎨 Image prompt

Rewrite the scene as a prompt for an image model.

`POST /image/prompt`

Every call can override `model` and `temperature`. No state, no database.

Mode 1 — the window view

In: a scene with a point of view, bearing, time of day, features.

```
{
  "geojson": {
    "pov": {"lat": 44.49, "lng": 11.34},
    "bearing": 90,
    "time_of_day": "07:30",
    "features": [
      {"properties": {
        "type": "buildings",
        "name": "market hall"}}
    ]
  },
  "detail_level": "medium"
}
```


Out: prose, no map-speak.

> *You're looking east into the early light. Just across the way the old market hall squares off against the morning, its long roofline catching the first of the sun while the street in front is still half in shade.*

DISTANCE IN HUMAN TERMS · VAGUE WHERE THE DATA IS

Mode 2 — ask the scene

Pass a `prompt` alongside the scene. The model reasons over **what's present** — and stays honest about what isn't.

```
{
  "geojson": { "...": "..." },
  "prompt": "What's the quickest
            way out on foot?"
}
```

GROUNDED · CONVERSATIONAL · 1–2 PARAGRAPHS

> *From here your fastest line is to keep the market hall on your left and follow the lane it fronts onto — it bends toward the open ground you can just make out*
> *ahead, away from the tighter streets behind you.*

Mode 3 — the image prompt

Same scene, retargeted for an **image-generation model** — architecture and city structure first, so the render is faithful to the *place*.

```
Ground-level view looking east at dawn down a cobbled market street.
A long stone market hall dominates the right, its arcaded façade and
pitched roof catching low golden light; narrow row houses recede on
the left into soft morning shade. Medieval European town, wide-angle,
photographic, soft volumetric light.
```

IMAGE_SYSTEM[] ADDS STYLE HINTS · DETAIL_LEVEL TUNES DENSITY

How it works

```
GeoJSON scene  ───▶  geobard  ───▶  prose / image-prompt
  features           mode-specific    { "text": "…",
  pov, bearing       system + user      "model": "…" }
  time_of_day        prompt → chat
  custom props       completion
```

- **No magic** — careful per-mode prompts do the translation, the model writes
- **Non-standard GeoJSON fields** are read as extra context, not ignored
- Pure functions over `(geojson, options, client)` — framework-free, reusable

Bring your own model

geobard speaks the **OpenAI chat-completions API** — so it runs against whatever you already use.

Default	OpenRouter (`https://openrouter.ai/api/v1`)
Swap	set `GEOBARD_OPENAI_BASE_URL`
Works with	OpenAI · Ollama · Groq · Together · any compatible endpoint
Per request	override `model` and `temperature`

LOCAL WITH OLLAMA · HOSTED WITH OPENROUTER · PICK PER CALL

The API surface

ROUTE	DOES
`POST /narrate/window`	Window view. `geojson`, `system[]`, `detail_level` → `{text, model}`
`POST /narrate/prompt`	Answer a `prompt` about the scene
`POST /image/prompt`	Image-gen prompt; `image_system[]` style hints
`GET /healthz`	`{"status":"ok"}` once env is valid (Docker healthcheck)

All bodies take optional `model` and `temperature`. Narration modes accept `system[]` prompt lines; window & image modes take `detail_level`.

Running in two commands

1 · start the service

```
docker run --rm -p 8000:8000 \
  -e GEOBARD_OPENAI_API_KEY=sk-or-… \
  -e GEOBARD_OPENAI_MODEL=\
anthropic/claude-haiku-4.5 \
  ghcr.io/openfantasymap/geobard:latest
```

2 · narrate a scene

```
curl -sS \
  localhost:8000/narrate/window \
  -H "content-type: application/json" \
  -d '{ "geojson": { … },
        "detail_level": "medium" }'
# → { "text": "…", "model": "…" }
```

IMAGE: GHCR.IO/OPENFANTASYMAP/GEOBARD · MIT OR APACHE-2.0

Configuration

ENV	REQUIRED	DEFAULT
`GEOBARD_OPENAI_API_KEY`	yes	—
`GEOBARD_OPENAI_MODEL`	yes	—
`GEOBARD_OPENAI_BASE_URL`	no	`openrouter.ai/api/v1`
`GEOBARD_TEMPERATURE`	no	`0.7`
`GEOBARD_HOST` / `GEOBARD_PORT`	no	`0.0.0.0` / `8000`

Fail-fast: `/healthz` touches settings, so a missing key surfaces immediately.

Where geobard lives: GaiaWM

GaiaWM is world infrastructure for AI — a simulation-intelligence platform that layers space, time, society and causal influence over 40+ canonical worlds.

🗺️ **Spatial**	40+ worlds, GIS precision — Toril, Star Wars, Middle-earth
⏰ **Temporal**	Time-aware state — history, seasons, cycles
👥 **Social**	Factions, relationships, reputation
🌊 **Influence**	Causal simulation — channels & propagation
🧠 **Narration → geobard**	Perception & description — geobard & EyeOnWorld

geobard & EyeOnWorld — one engine

> *Click any location on the map — get an immersive description of standing there.*

EyeOnWorld (live on **openfantasymaps.org**) is the same narration — geobard's window view — wired to real world data:

- Map tile → GeoJSON scene → `/narrate/window`
- The same `/image/prompt` mode renders the view
- Agents use `/narrate/prompt` **to reason over a place**

Looking around, you're standing at the edge of a residential neighborhood where
the midday sun beats down on the cobblestones. Selduth Street is just a few paces to your right...

EyeOnWorld · generated from Waterdeep map data

Integrate it

- **As a service** — `docker run`, POST your scenes, get text back
- **As a library** — import the pure helpers from `geobard.llm` directly
- **In an agent loop** — `/narrate/prompt` gives spatial grounding over a scene
- **In a pipeline** — `/image/prompt` → your image model of choice
- **Self-hosted & private** — point the base URL at a local Ollama

Small surface, few dependencies — FastAPI, Pydantic, the OpenAI client.

GEOBARD

A scene goes in. *Words* come out.

The open-source narration engine from GaiaWM — the engine behind EyeOnWorld.

github.com/openfantasymap/geobard · MIT or Apache-2.0

gaiawm.github.io · openfantasymaps.org · github.com/openfantasymap/geobard